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NMAM INSTITUTE OF TECHNOLOGY, NITTE

(An Autonomous Institution affiliated to VTU, Belagavi)

Fourth Semester B.E. (Civil/Mechanical) (Credit System) Degree Examinations

Supplementary Examinations - December 2025

21CV401 / 20CV401 / 21ME401 / 20ME401 - ENGINEERING MATHEMATICS - IV

Duration: 3 Hours

Max. Marks: 100

Note: 1) Answer **Five full** questions choosing **One full** question from **Unit – I** and **Two full** questions from **Unit – II & Unit – III** each.

2) **Statistical tables** may be provided.

Unit – I

Marks BT* CO* PO*

1. a) Find the value of x at $y = 15$ using inverse Lagrange's interpolation formula from the following data:

x	5	6	9	11
y	12	13	14	16

6 L*1 1 2

- b) A function $y = f(x)$ is given by the following table.

x	1.0	1.2	1.4	1.6	1.8	2.0
y	0	0.128	0.544	1.296	2.432	4.00

Find $\frac{dy}{dx}$ and $\frac{d^2y}{dx^2}$ at 1.0 by suitable interpolation formula.

7 L1 1 2

- c) Using Newton's forward difference interpolation formula, compute the pressure of the steam at the temperature of 142° from the steam table.

Temperature	140	150	160	170	180
Pressure	3.685	4.854	6.302	8.076	10.225

7 L2 1 2

2. a) Evaluate $\int_0^1 \frac{1}{1+x^2} dx$ by using Simpson's $\frac{1}{3}$ rd rule taking four equal strips.

6 L1 1 2

- b) Derive the general quadrature formula and hence deduce Trapezoidal rule.

7 L2 1 1

- c) Using Newton's divided difference formula, find $f(8)$ from the following data.

x	4	5	7	10	11	13
$f(x)$	48	100	294	900	1210	2028

7 L1 1 2

Unit – II

3. a) Using Regula-Falsi method find an approximate root of the equation $x^3 - 2x - 5 = 0$ in the interval (2, 3). Carry out three iterations.

6 L2 2 2

- b) Using modified Euler's method, find an approximate value of y when $x = 0.1$ given that $\frac{dy}{dx} = x - y^2$ and $y(0) = 1$. Take $h = 0.1$ and carry out two iterations.

7 L1 2 2

- c) Solve the PDE $\nabla^2 u = -10(x^2 + y^2 + 10)$ over the square with sides $x = 0 = y$; $x = 3 = y$ with $u = 0$ on the boundary and $h = 1$.

7 L3 3 2

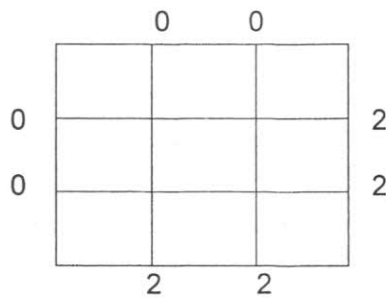
4. a) Using Newton-Raphson method find an approximate root of the equation $x^3 - 2x - 11 = 0$ near $x_0 = 2$. Carry out three iterations.

6 L2 2 2

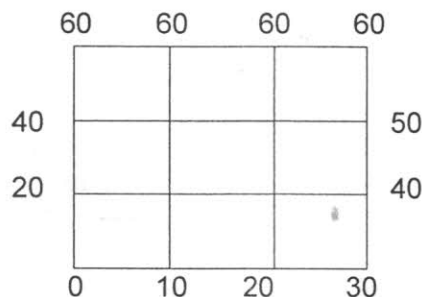
- b) Find the solution of the equation $u_{xx} = 2u_t$ with $u(0, t) = u(4, t) = 0$ and $u(x, 0) = x(4 - x)$, taking $h = 1, k = 1$. Find the values up to $t = 3$.

7 L1 3 2

- c) Solve $u_{xx} + u_{yy} = 0$ for the square mesh with the boundary values as shown below:



5. a) Using Taylor's series method of order four compute y at $x = 0.1$ given $\frac{dy}{dx} = 2y + 3e^x$, $y(0) = 0$, taking $h = 0.1$.
 b) Using fourth order Runge-Kutta method, solve the initial value problem $\frac{dy}{dx} = x + y$, $y(0) = 1$, at $x = 0.2$ taking $h = 0.2$.
 c) Solve $u_{xx} + u_{yy} = 0$ for the square mesh with the boundary values as shown below:



Unit – III

6. a) If a random variable X has an exponential distribution with a mean of $\frac{1}{2}$. Find the probability that (i) X will lie between 1 and 3
 ii) X is greater than 0.5.
 b) State and prove Bayes' theorem.
 c) The probability distribution of a random variable is given by the following table:

X	0	1	2	3	4	5
P(X)	k	$5k$	$10k$	$10k$	$5k$	k

Find (i) the value of k (ii) the mean (iii) $P(X < 3)$.

7. a) The probability that an individual suffers a bad reaction from an injection is 0.001. Using Poisson distribution, determine the probability that out of 2000 individuals (i) exactly 3 (ii) more than 1 individual suffers a bad reaction.
 b) Obtain the mean and variance of a Poisson distribution.
 c) A random variable X has the density function

$$f(x) = \begin{cases} k(1 - x^2) & , 0 < x < 1 \\ 0 & , \text{elsewhere} \end{cases}$$

Find (i) the value of k (ii) $P(0.1 < X < 0.2)$ (iii) $P(X > 0.5)$.

8. a) Mr. Sharma who lives in the outskirts of a city wants to catch an early morning train. The probability that he gets an auto at that time is only 0.23. If he gets an auto, the probability of catching the train is 0.85. If he fails to get an auto, he has to take a city bus and probability he gets train is only 0.43. Find the probability that he catches the train.

7 L3 3 2

6 L1 2 2

7 L3 2 2

7 L2 3 2

6 L1 5 2

7 L2 4 1

7 L1 4 2

6 L1 5 2

7 L2 5 1

7 L1 4 2

6 L1 4 2

- b) A joint distribution of (X, Y) is given by the following table.

X \ Y	1	2	3
1	$\frac{1}{12}$	$\frac{1}{6}$	0
2	0	$\frac{1}{9}$	$\frac{1}{5}$
3	$\frac{1}{18}$	$\frac{1}{4}$	$\frac{2}{15}$

Determine (i) marginal distributions of X & Y (ii) Test the independence of X & Y .

- c) Suppose that X has distribution $N(2, 0.16)$. Evaluate

(i) $P(X \geq 2.3)$ (ii) $P(1.8 \leq X \leq 2.1)$.

7 L1 4 2

7 L2 5 2

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
